

1. Project Title : Vendor Performance Analysis

2. Business Problem

Retail and wholesale businesses depend heavily on effective vendor management, inventory optimization, and pricing strategies to maximize profitability. Inefficient purchasing decisions, excessive inventory, and over-reliance on a small number of vendors can negatively impact business performance and cash flow.

This project aims to analyze vendor performance using sales, purchase, pricing, and inventory data to identify opportunities for cost optimization and profit improvement.

Key Objectives:

- Identify underperforming brands with high profit margins but low sales that may benefit from targeted promotions or pricing adjustments.
- Determine the vendors and brands contributing the highest sales and profitability.
- Evaluate vendor dependency by analyzing purchase contribution across suppliers.
- Analyze the impact of bulk purchasing on unit costs and identify opportunities for procurement savings.
- Identify vendors with low inventory turnover and excess stock levels.
- Quantify capital locked in unsold inventory and determine the vendors contributing most to inventory holding costs.
- Use statistical techniques to compare profitability patterns between high-performing and low-performing vendors and validate findings through hypothesis testing.

3. Dataset Overview

The project uses six operational datasets containing sales, purchase, inventory, pricing, and vendor information.

<u>Dataset</u>	<u>Description</u>	<u>Rows</u>
sales.csv	Product-level sales transactions	12,825,363
purchases.csv	Product purchase transactions	2,372,474
begin_inventory.csv	Opening inventory records	206,529
end_inventory.csv	Closing inventory records	224,489
purchase_prices.csv	Product purchase pricing details	12,261
vendor_invoice.csv	Vendor invoice and freight information	5,543

Total Records Processed: 15.6+ Million

The raw CSV files were loaded into a SQLite database using a custom ETL pipeline built in Python. Data was processed in chunks of 50,000 records to efficiently handle large datasets and optimize memory usage. The datasets

were then integrated to create a consolidated vendor performance table for analysis.

ETL Process

A Python-based ETL pipeline was developed to automate data ingestion and database creation. The pipeline:

- Loaded large CSV files into SQLite using chunk processing.
- Automated table creation and data loading.
- Logged execution status, processing time, row counts, and errors.
- Created a centralized database for analysis and reporting.

Technologies Used

- Python
- Pandas
- SQLite
- SQLAlchemy
- Power BI

Business Areas Covered

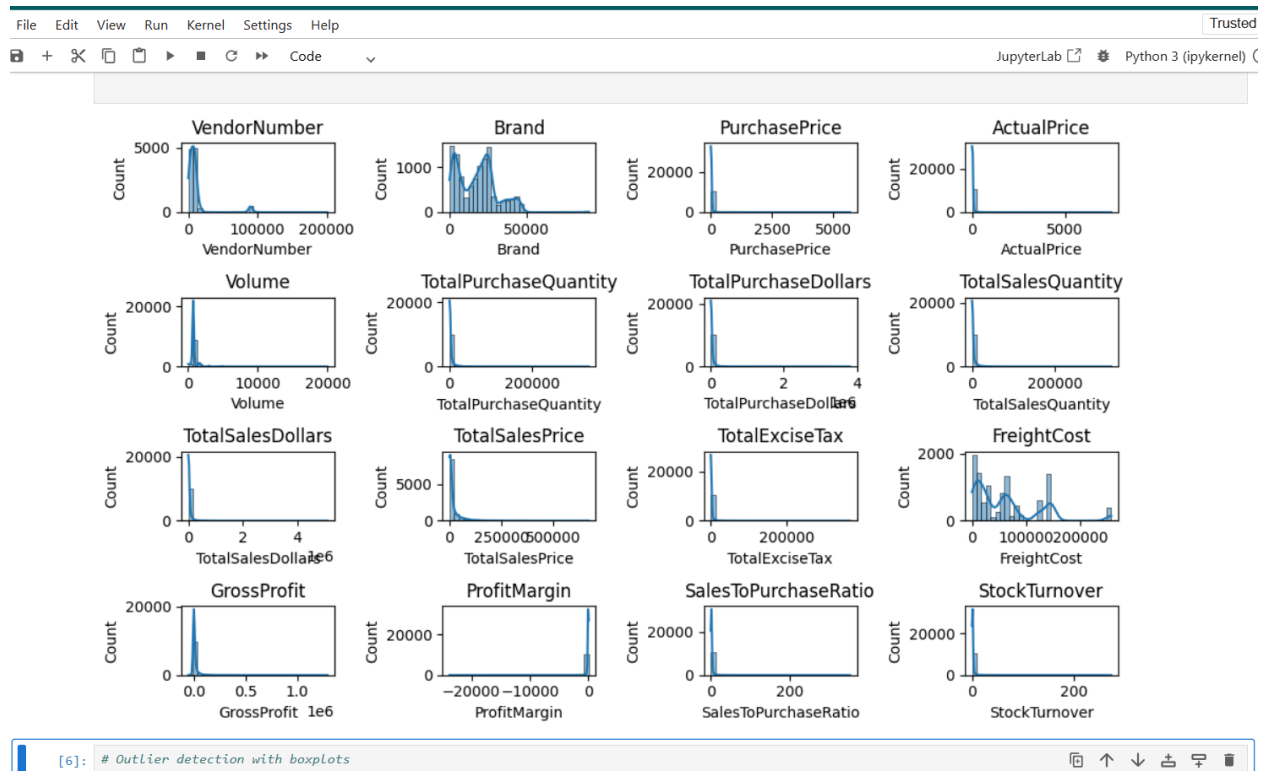
- Vendor Performance Analysis
- Brand Performance Evaluation
- Sales Analysis
- Inventory Management
- Purchase Cost Analysis
- Profitability Assessment

4. Exploratory Data Analysis Insights

Summary statistics

[4]:

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02



Negative and Zero Values

Gross Profit:

- The minimum Gross Profit value is **-52,002.78**, indicating that certain products or vendor-brand combinations generated losses.
- This may occur when purchase and operational costs exceed sales revenue or when products are sold at heavily discounted prices.

Profit Margin:

- The minimum Profit Margin is **negative infinity** ($-\infty$), which occurs when Total Sales Dollars equals zero while costs have been incurred.
- This suggests the presence of products that were purchased but generated no sales revenue during the analysis period.

Sales Quantity and Sales Revenue:

- Minimum values of **0** for Total Sales Quantity and Total Sales Dollars indicate that some purchased products were never sold.
- These products may represent slow-moving inventory, excess stock, or items with low customer demand.

Outlier Analysis

Purchase Price and Actual Price:

- Maximum values (**5,681.81** and **7,499.99**) are substantially higher than their respective mean values (**24.39** and **35.64**).
- This indicates the presence of premium or luxury products that are priced significantly higher than the majority of products in the dataset.

Freight Cost:

- Freight Cost ranges from **0.09** to **257,032.07**, showing considerable variation in logistics expenses across vendors.
- High freight costs may be associated with bulk orders, large shipment volumes, or differences in supplier locations.

Stock Turnover:

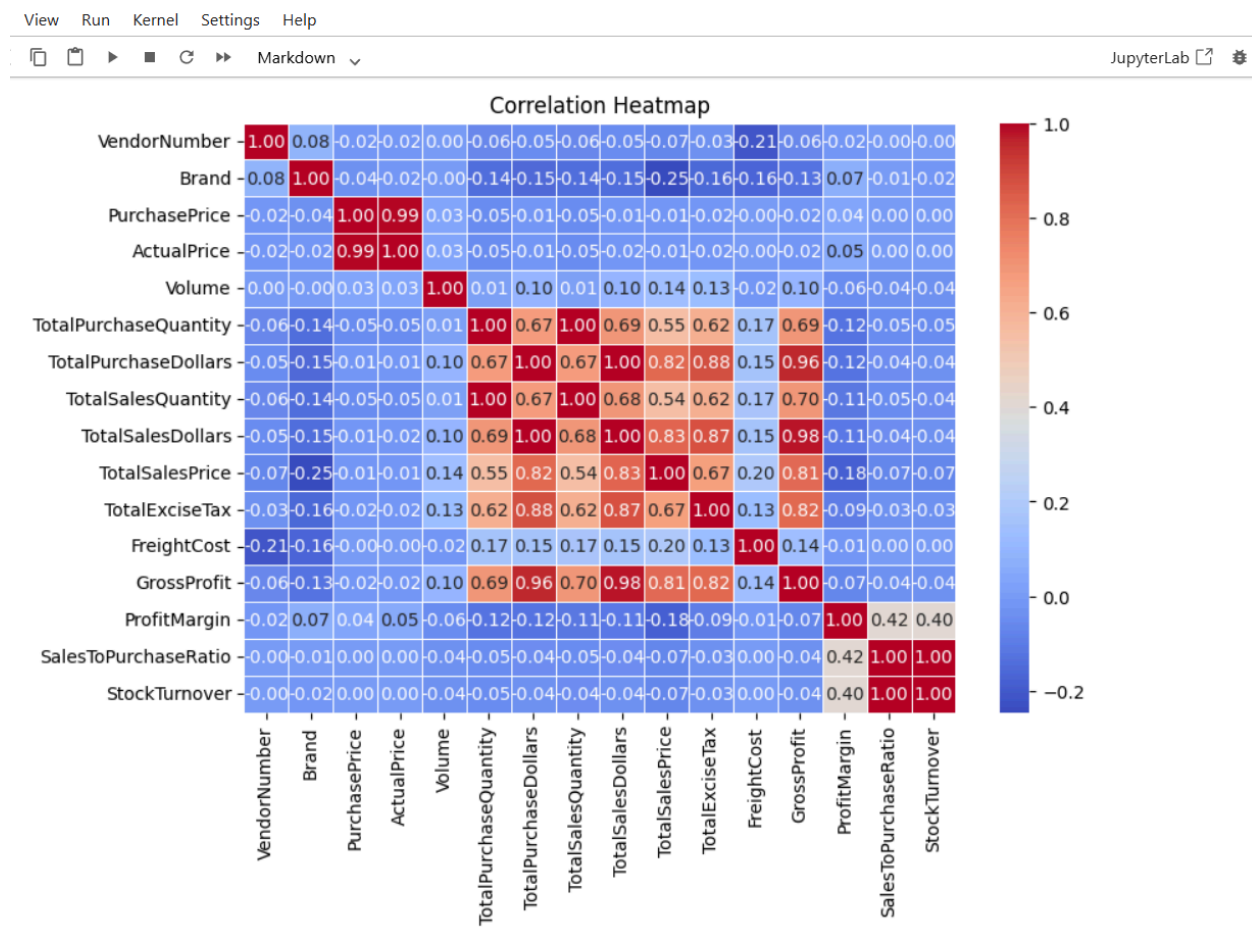
- Stock Turnover ranges from **0** to **274.5**, highlighting substantial differences in product movement.
- A value of **0** indicates products that were purchased but not sold.
- Values greater than **1** suggest that sales quantity exceeded purchases during the analysis period, likely because sales were fulfilled using inventory carried over from previous stock periods.
- Very high turnover values indicate fast-moving products with strong market demand.

Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions)
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Correlation Insights



Purchase Price vs Sales Revenue and Profit

- Purchase Price shows a very weak negative correlation with Total Sales Dollars (-0.012) and Gross Profit (-0.016).

- This indicates that product purchase cost has little direct relationship with overall sales revenue or profitability in the dataset.
- Factors such as sales volume, product demand, and pricing strategy may have a greater influence on business performance.

Purchase Quantity vs Sales Quantity

- A very strong positive correlation (0.999) exists between Total Purchase Quantity and Total Sales Quantity.
- This suggests that products purchased in larger quantities generally experience higher sales volumes, indicating effective inventory movement and demand alignment.

Profit Margin vs Total Sales Price

- Profit Margin has a weak negative correlation (-0.179) with Total Sales Price.
- This suggests that higher-priced products do not necessarily generate higher profit margins.
- The result may indicate that some premium products carry higher acquisition costs, reducing overall margin percentages.

Stock Turnover vs Profitability

- Stock Turnover shows weak negative correlations with Gross Profit (-0.038) and Profit Margin (-0.055).
- This indicates that faster inventory movement alone does not guarantee higher profitability.
- Product pricing, operating costs, and vendor-related expenses also play important roles in determining profit performance.

5. Research Questions and key Findings

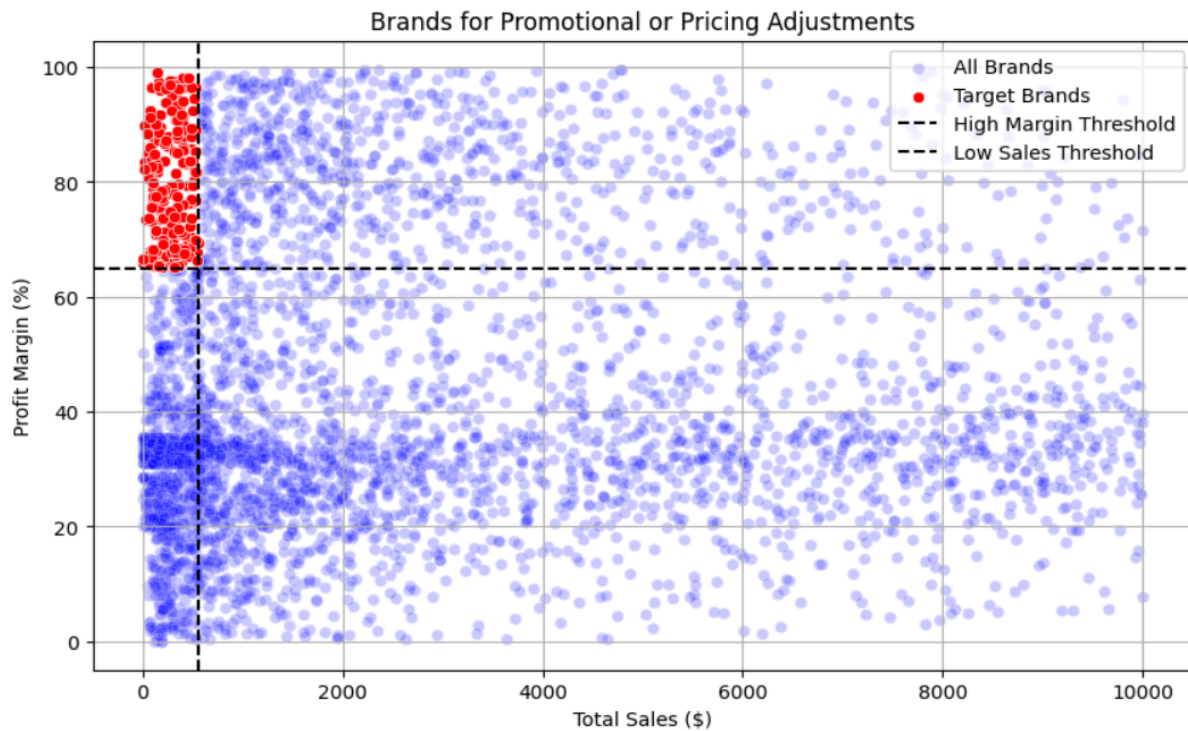
1. Brands for Promotional or Pricing Adjustments

Brands with low sales but high Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournal 10	558.87	69.434752

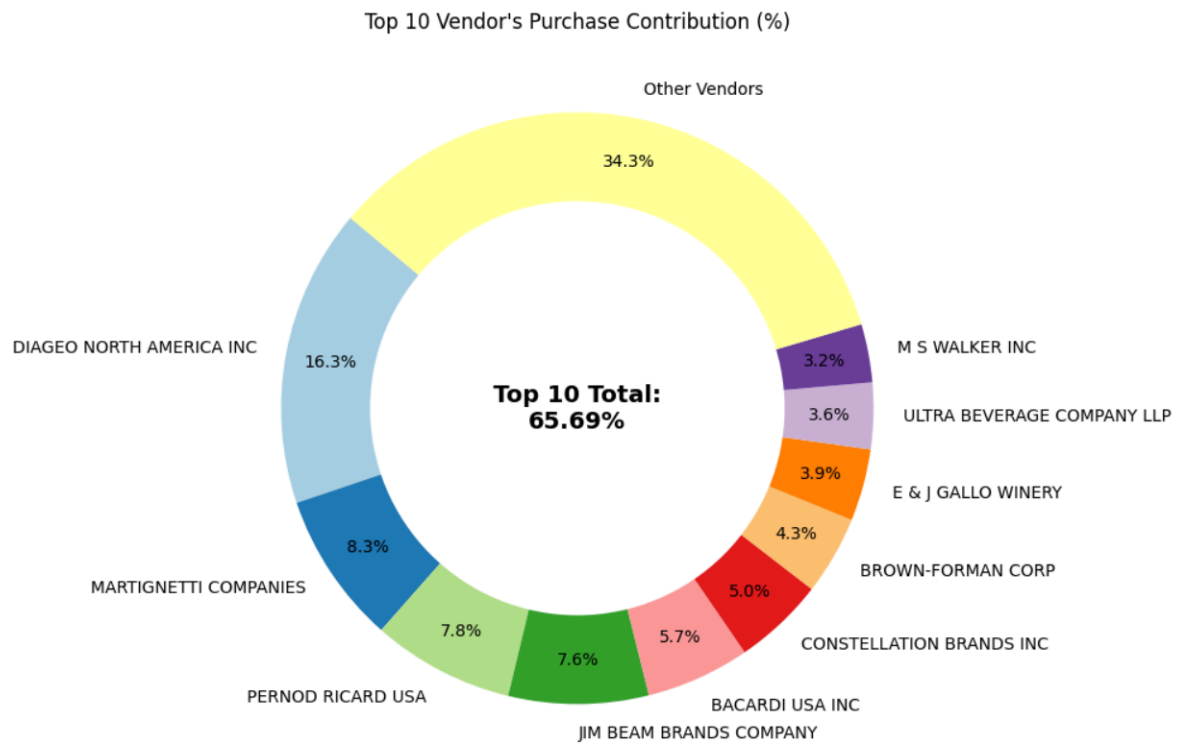
198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins which could benefit from targeted marketing, promotion, or price optimizations to increase volume without compromising profitability.



2. Top Vendors by Sales and Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

OrderSize	UnitPurchasePrice
Small	39.068186
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: \$2.17M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

VendorName	UnsoldInventoryValue
DIAGEO NORTH AMERICA INC	722.21K
JIM BEAM BRANDS COMPANY	554.67K
PERNOD RICARD USA	470.63K
WILLIAM GRANT & SONS INC	401.96K
E & J GALLO WINERY	228.28K
SAZERAC CO INC	198.44K
BROWN-FORMAN CORP	177.73K
CONSTELLATION BRANDS INC	133.62K
MOET HENNESSY USA INC	126.48K
REMY COINTREAU USA INC	118.60K

5. Profit Margin Comparison: High vs. Low-Performing Vendors

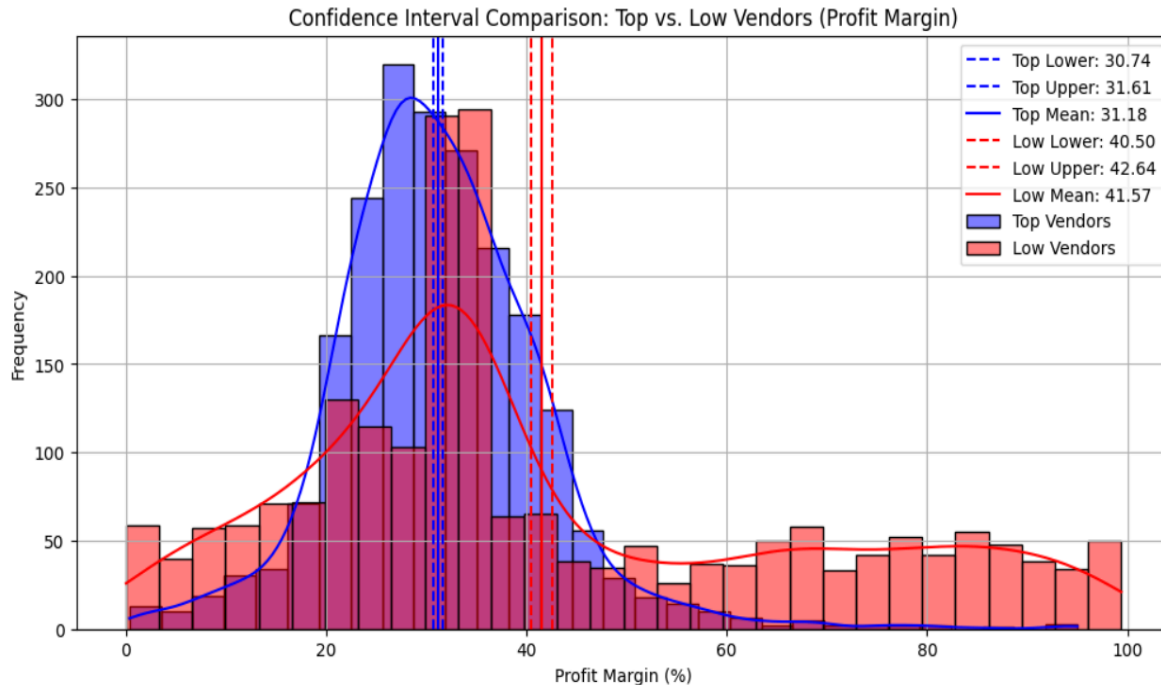
Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%

Low Vendors' Profit Margin (95% CI): (40.48%, 62.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



6. Statistical Validation of Profit Margin Differences

Hypothesis Testing:

H_0 (Null Hypothesis):

No significant difference in profit margins exists between top-performing and low-performing vendors.

H_1 (Alternative Hypothesis):

A significant difference exists in profit margins between the two vendor groups.

Result:

The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication:

High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on improving cost efficiency.

Final Recommendations

- Re-evaluate pricing for low sales, high-margin brands to boost sales volume without sacrificing profitability.

- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.